

Gradient Descent in ML.

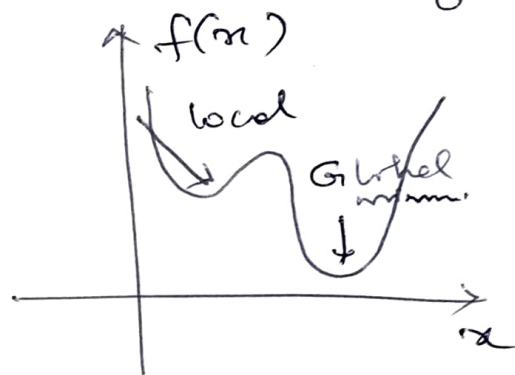
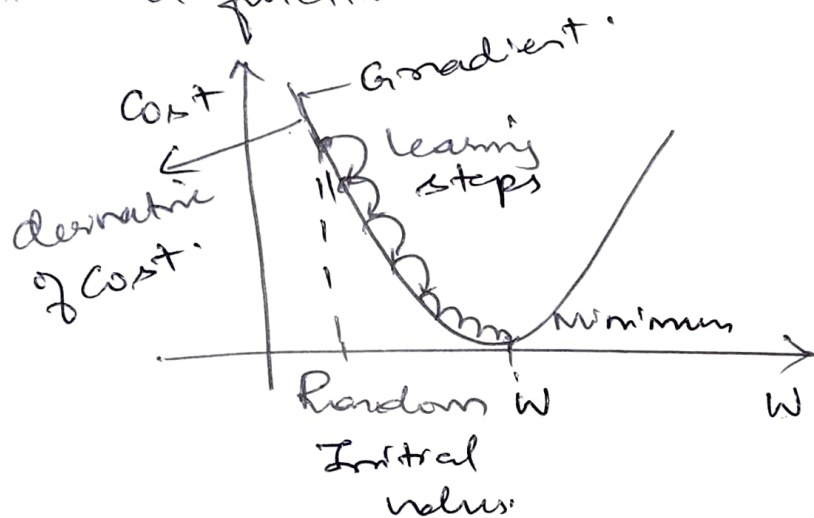
①

* Most Commonly used optimization algorithm to train machine learning models by means of minimizing errors b/w actual and expected results

* Train Neural N/w's.

* objective is to minimize the convex function using iteration of parameter update.

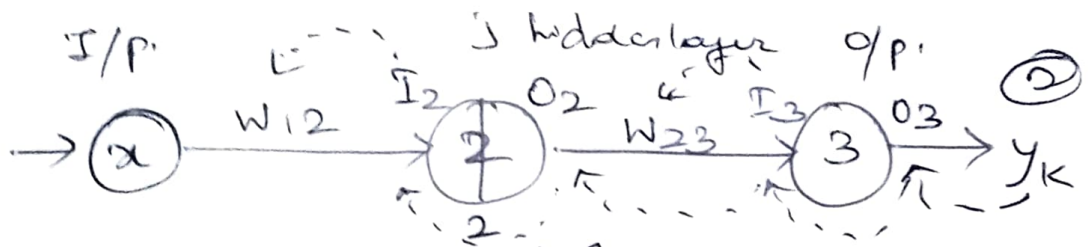
* helps in finding the local minimum of a function



* Moving towards a negative gradient (α) away from the gradient of function at the current point.

① Calculate the first-order derivative of the function to compute the gradient (α), slope of that function

② Move away from the gradient, which means the direction of the slope increased from the current point by ' α ' time where ' α ' - learning rate.



$$I_2 = W_{12} x \quad O_2 = \phi(I_2) \quad \text{Target: } Y$$

$$I_3 = O_2 W_{23} \quad O_3 = \phi(I_3) \quad \text{Output: } O_3$$

Cost function: $J(w) = \frac{1}{2} (Y - O_3)^2$ [Mean squared Error]

how could we express the relationship b/w $J(w)$ with weights.

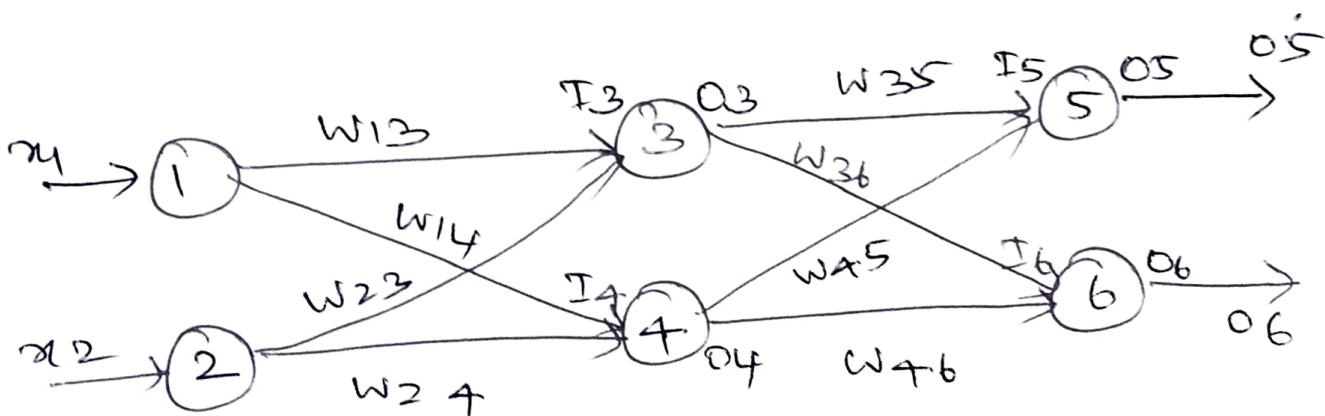
$$\frac{\partial J(w)}{\partial w_2} = ? \quad \& \quad \frac{\partial J(w)}{\partial w_1} = ? \quad \text{weights.}$$

$$\frac{\partial}{\partial z} P(q(z)) = \frac{\partial P}{\partial q} \frac{\partial q}{\partial z}$$

$$\frac{\partial J(w)}{\partial w_{23}} = \frac{\partial J(w)}{\partial O_3} \frac{\partial O_3}{\partial I_3} \frac{\partial I_3}{\partial w_{23}}$$

By similar logic

$$\frac{\partial J(w)}{\partial w_{12}} = \frac{\partial J(w)}{\partial O_3} \frac{\partial O_3}{\partial I_3} \frac{\partial I_3}{\partial O_2} \frac{\partial O_2}{\partial I_2} \frac{\partial I_2}{\partial w_{12}}$$



Cost function:

$$J(w) = \frac{1}{2m} \sum (y_i - o_k)^2$$

Impact of weight w_{13}

(3)

$$\frac{\partial I(w)}{\partial w_{13}} = \left\{ \frac{\frac{\partial I(w)}{\partial o_5}}{\frac{\partial I_5}{\partial o_3}} \frac{\partial I_5}{\partial o_3} \frac{\partial o_3}{\partial I_3} \frac{\partial I_3}{\partial w_{13}} \right\} + \left\{ \frac{\frac{\partial I(w)}{\partial o_6}}{\frac{\partial I_6}{\partial o_3}} \frac{\partial I_6}{\partial o_3} \frac{\partial o_3}{\partial I_3} \frac{\partial I_3}{\partial w_{13}} \right\}$$

Impact of weight w_{35}

$$\frac{\partial I(w)}{\partial w_{35}} = \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial w_{35}}$$

$$\frac{\partial I(w)}{\partial w_{36}} = \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial w_{36}}$$

$$\frac{\partial I(w)}{\partial w_{45}} = \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial w_{45}}$$

$$\frac{\partial I(w)}{\partial w_{46}} = \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial w_{46}}$$

Parameter of Interest changes w.r. to weighted inputs.

Generalization

Partial derivative of Cost function

derivative of Activation function.

$$\frac{\partial I(w)}{\partial o_6} = \frac{1}{m_i} (y_i - o_6) f'(\phi_k) \frac{\partial o_5}{\partial I_5} = \frac{f'(\phi_5)}{f'(\phi_5)}$$

$$\delta^i = \frac{1}{m} (y_i - o_k) f'(\phi_k)$$

$$\frac{\partial I(w)}{\partial w_{35}} = \delta_5 \left(\frac{\partial I_5}{\partial w_{35}} \right)$$

④

$$\frac{\partial I(w)}{\partial w_{45}} = \delta_5 \left(\frac{\partial I_5}{\partial w_{45}} \right)$$

$$\frac{\partial I(w)}{\partial w_{36}} = \delta_6 \left(\frac{\partial I_6}{\partial w_{36}} \right)$$

$$\frac{\partial I(w)}{\partial w_{46}} = \delta_6 \left(\frac{\partial I_6}{\partial w_{46}} \right)$$

The last derivative term ends up representing activation from previous layer.

$$\frac{\partial I(w)}{\partial w_{35}} = \delta_5 \cdot 0.3 \quad \frac{\partial I(w)}{\partial w_{46}} = \delta_2 \cdot 0.4$$

$$\frac{\partial I(w)}{\partial w_{45}} = \delta_1 \cdot 0.4 \quad \frac{\partial I(w)}{\partial w_{36}} = \delta_2 \cdot 0.3$$

$$\frac{\partial I(w)}{\partial w_{ij}} = \begin{bmatrix} \delta_i \\ \delta_j \end{bmatrix} \begin{bmatrix} 0.3 & 0.4 \end{bmatrix}$$

$i = 3, 4$
 $j = 5, 6$

error term = 8

$$\text{Input} = \frac{\partial I_6}{\partial w_{36}} \quad (0.3/0.4)$$

Layer 1 Parameters

(5)

$$\frac{\partial I(w)}{\partial w_{13}} = \left\{ \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial o_3} \frac{\partial I_3}{\partial w_{13}} \right\} + \left\{ \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial o_3} \frac{\partial I_3}{\partial w_{13}} \right\}$$

Input.

$$\frac{\partial I(w)}{\partial w_{14}} = \left\{ \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial o_4} \frac{\partial I_4}{\partial w_{14}} \right\} + \left\{ \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial o_4} \frac{\partial I_4}{\partial w_{14}} \right\}$$

$$\frac{\partial I(w)}{\partial w_{23}} = \left\{ \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial o_3} \frac{\partial I_3}{\partial w_{23}} \right\} + \left\{ \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial o_3} \frac{\partial I_3}{\partial w_{23}} \right\}$$

$$\frac{\partial I(w)}{\partial w_{24}} = \left\{ \frac{\partial I(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial o_4} \frac{\partial I_4}{\partial w_{24}} \right\} + \left\{ \frac{\partial I(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial o_4} \frac{\partial I_4}{\partial w_{24}} \right\}$$

example:

$$\begin{aligned} \frac{\partial I(w)}{\partial w_{13}} &= \delta_5 \left(\frac{\partial o_5}{\partial I_5} \right) \left(\frac{\partial I_5}{\partial o_3} \right) \left(\frac{\partial I_3}{\partial w_{13}} \right) + \delta_6 \left(\frac{\partial I_6}{\partial o_3} \right) \left(\frac{\partial o_3}{\partial I_3} \right) \left(\frac{\partial I_3}{\partial w_{13}} \right) \\ &= \delta_5 w_{35} f'(\phi_3) \left(\frac{\partial I_3}{\partial w_{13}} \right) + \delta_6 w_{36} f'(\phi_3) \left(\frac{\partial I_3}{\partial w_{13}} \right) \end{aligned}$$

$$\frac{\partial \mathcal{I}(\omega)}{\partial w_{13}} = \delta_5 w_{35} f'(\phi_3) \left(\frac{\partial \mathcal{I}_3}{\partial w_{13}} \right) + \quad (6)$$

$$\delta_6 w_{36} f'(\phi_3) \left(\frac{\partial \mathcal{I}_3}{\partial w_{13}} \right)$$

$$= \frac{\partial \mathcal{I}_3}{\partial w_{13}} \left[\delta_5 w_{35} f'(\phi_3) + \delta_6 w_{36} f'(\phi_3) \right]$$

$$\boxed{\frac{\partial \mathcal{I}_3}{\partial w_{13}} = \delta_3 \pi_1}$$

$$\frac{\partial \mathcal{I}(\omega)}{\partial w_{14}} = \left\{ \frac{\partial \mathcal{I}(\omega)}{\partial o_5} \frac{\partial o_5}{\partial \mathcal{I}_5} \frac{\partial \mathcal{I}_5}{\partial o_4} \frac{\partial o_4}{\partial \mathcal{I}_4} \frac{\partial \mathcal{I}_4}{\partial w_{14}} \right\} + \left\{ \frac{\partial \mathcal{I}(\omega)}{\partial o_6} \frac{\partial o_6}{\partial \mathcal{I}_6} \frac{\partial \mathcal{I}_6}{\partial o_4} \frac{\partial o_4}{\partial \mathcal{I}_4} \frac{\partial \mathcal{I}_4}{\partial w_{14}} \right\}$$

$$= \delta_5 w_{45} f'(\phi_4) \frac{\partial \mathcal{I}_4}{\partial w_{14}} + \delta_6 w_{46} f'(\phi_4) \frac{\partial \mathcal{I}_4}{\partial w_{14}}$$

$$= \frac{\partial \mathcal{I}_4}{\partial w_{14}} \left[\delta_5 w_{45} f'(\phi_4) + \delta_6 w_{46} f'(\phi_4) \right]$$

$$= \frac{\partial \mathcal{I}_4}{\partial w_{14}} [\delta_4] = \pi_1 \delta_4$$

In General term

$$\frac{\partial J(\omega)}{\partial w_{ij}^{po}} = \left(\delta_j^p \cdot \overset{O_i}{w_{ji}^{th}} \right)$$

example

$$\frac{\partial J(\omega)}{\partial w_{4b}} = \delta_b^o \cdot 4_i$$

Backward propagation